DOST Form 2 (for Basic/Applied Research)

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(1) PROJECT PROFILE

Program Title:	None
Project Title:	**Development of Vegan Leather from Banana (*Musa acuminata* × *balbisiana* var. Cardaba) Peels Reinforced with Snake Plant (*Sansevieria trifasciata*) Fibers**
Project Leader/Sex:	Engr. Kier P. Gasga/Male
Project Duration (number of months):	12 months
Project Start Date:	July 2022
Project End Date:	June 2023
Implementing Agency (Name of University-College-Institute, Department/Organization or Company):	Partido State University
Address/Telephone/Fax/Email (Barangay, Municipality, District, Province, Region):	San Juan Evangelista Street, Goa, Camarines Sur kier.gasga@parsu.edu.ph/ op@parsu.edu.ph +639459734519

(2) COOPERATING AGENCY/IES (Name/s and Address/es)

None

(3) SITE(S) OF IMPLEMENTATION

IMPLEMENTATION SITES NO.	COUNTRY	REGION	PROVINCE	DISTRICT	MUNICIPALITY	BARANGAY
1.	Philippines	V (Bicol)	Camarines Sur	4th Legislative District (Partido)	San Jose	Del Carmen
2.	Philippines	V (Bicol)	Camarines Sur	4th Legislative District (Partido)	Goa	San Juan Evangelista St.
3.	Philippines	V (Bicol)	Camarines Sur	4th Legislative District (Partido)	Lagonoy	Burabod, Sta. Maria & San Ramon

(4) TYPE OF RESEARCH

_____ Basic
__✓__ Applied

(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

_____ Agriculture, Aquatic and Natural Resources
Commodity: _____
_____ Health
Priority Topic: _____
__✓__ Industry, Energy and Emerging Technology
Sector: Process-Textile and Agriculture (Banana & Natural Fiber Crops) & Environment *(from BCIEERD R&D Agenda*
_____ Disaster Risk Reduction and Climate Change Adaptation
_____ Basic Research
Sector: _____

Sustainable Development Goal (SDG) Addressed







(6) EXECUTIVE SUMMARY (not to exceed 200 words)

The leather industry is considered as one of the most polluting industry due to high carbon footprint and the contamination brought by the tanning process. Moreover, synthetic leather that are plastic-based are not biodegradable and can release dangerous toxins. Plant-based leather is seen as an alternative to this product. This material, although considered as a niche category, is rapidly increasing and its demand is accelerated by veganism and international-brand adopters. On the other hand, banana (*Musa acuminata* × *balbisiana* var. Cardaba) peel is considered as typical waste after consumption and when disposed by local vendors and banana crisp industries. This waste contributes to disposal problems and may end up in landfills which require waste treatment. In this proposed project, banana peels are reinforced with fibers from highly abundant snake plant (*Sansevieria trifasciata*) to develop a novel vegan leather. In addition, process on how to fabricate said product will be established and fabricated product will be evaluated. Test

sample products using the vegan leather will also be developed.

(7) INTRODUCTION

Conventional leather is one of fashion's most ubiquitous materials – but it is fraught with ethical and environmental issues. In fact, the production of leather is considered as one of the most polluting industries. Environmentally friendly and animal free leather also known as "vegan leather" is an alternative biomaterial produced without the use of any animal component. This biobased material compared to traditional leather shows similar physico-chemical and mechanical properties. Moreover, recent studies show that this class of materials are gradually gaining market in the fashion industry as leather substitutes.

On the other hand, bananas (*Musa acuminata* × *balbisiana* var. Cardaba) are considered as one of the most important tropical fruits in the world market. In the Philippines, banana is the most popular cultivar grown domestically which accounts to 39% of total production. Saba banana is most widely processed into chips/crackers which account for 95% of the country's earnings on processed banana. Banana fruit peels comprise a significant quantity of wastes produced from banana processing, which is equivalent to 40% of the total weight of fresh banana. These peels are just left as solid waste at large expense and are not being used for any other purposes. Thus, the operation of food processing wastes in the industry is now becoming a very serious ecological problem and additional pollution for the environment.

In addressing the concern with this waste, the project aims to utilize the banana peel as raw and main material for the fabrication of a vegan leather. Moreover, to increase the strength of the material, it will be reinforced with fibers derived from snake plant (*Sansevieria trifasciata*) which is a known ornamental plant that is highly abundant and is easy to propagate. This project not only aims to valorize banana peel but also pave way into creating a sustainable product.

(7.1) RATIONALE/SIGNIFICANCE (not to exceed 300 words)

Banana is one of the most common crops grown in tropical countries, including the Philippines. Its variety, cardaba or saba (*Musa acuminata* × *balbisiana* var. Cardaba) is considered to be the most important in the country due to its dominance in chips production and in other domestic and exported food products. Nevertheless, its peel, which constitute to 40% w/w of the banana fruit is often discarded as waste. Its properties which include its fiber content, however, reinforced with fibers from snake plant (*Sansevieria trifasciata*) proves its viability to be processed into vegan leather. With this, the proposed project would significantly address two problems: First, it would develop a plant-based vegan leather that would provide an answer to the environmental concerns brought by production of animal leather and plastic-based faux leather. And second, the project would develop a novel method in developing vegan leather using the mentioned main materials, hence valorizing banana peel.

(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK

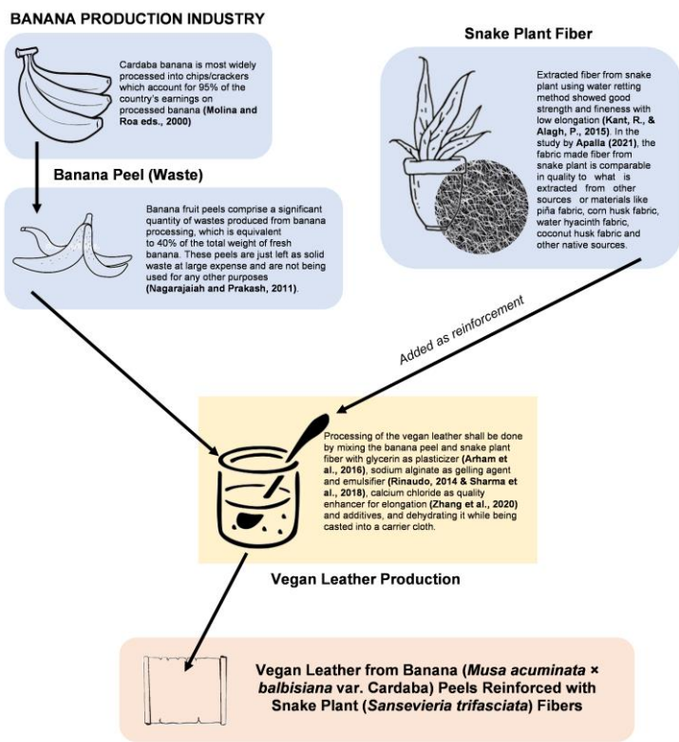


Fig 1. Framework of the Study

Figure 1 shows the framework of the study along with the scientific basis of the study. A detailed discussion is presented in the methodology.

(7.3) OBJECTIVES

General: The research project aims to develop a sustainable vegan leather from discarded banana (*Musa acuminata* × *balbisiana* var. Cardaba) peels reinforced with snake plant (*Sansevieria trifasciata*) fibers.

Specific: Specifically, the project aims to:

a. Establish a method of fabricating vegan leather using banana peels reinforced with snake plant fibers.
b. Evaluate the quality of the developed vegan leather; and
c. Develop representative test products using the developed vegan leather.

(8) REVIEW OF LITERATURE

Downside of Leather Industry

The leather making industry has been categorized as one of the highly polluting industries (Kanagaraj et al, 2006). The harmful nature of the chemicals used in the tanning process contaminates the environment. High contents of dissolved, suspended organic and inorganic solids in tannery effluents give rise to potentially toxic metal salts, chromium, and metal ions and increases biological oxygen demand. Moreover, tanning activities also emanate an unpleasant odor from the decomposition of protein waste material and the presence of sulfide, ammonia and other volatile compounds in the process. The salt used for preserving the hide discharges a large amount of total dissolved solids (TDS) and chlorides. There is an average consumption of 45-50 m^2 of the waste liquor and 800 kg of solid wastes per ton of rawhide discharged by the leather industry (Ozgunay et al., 2007). Other polluting chemicals in the tanning industry are lime, sodium sulfide, ammonium salts, sulfuric acid, chromium salts, and vegetable tanning materials. Solid wastes generated in the leather industry pose a pollution threat in terms of BOD and TDS. In addition, many leather industries dispose unprocessed water onto soil and affects the soil's productivity (Kanagaraj et al, 2006). These chemicals used in the making of leather impose a significant level of contamination on the environment.

Not only does the leather industry release chemicals that can give rise to potentially toxic metal salts and increases BOD, it is also linked with the expansion of animal cruelty because the sale of leather good is directly associated with the economic success of animal factory farms (PETA, 2018). Every year, the global leather industry tans more than a billion of byproducts (skins and hides) from the meatpacking industry. During the transport of the meat manufacturing, many of these animals endure extreme crowding and confinement, deprivation, branding, tail-docking, and dehorning (PETA, 2019). Although most leathers are made from domestic animals, other exotic species are also hunted and killed specifically for their skins, including zebras, bison, kangaroos, elephants, crocodiles, alligators, ostriches, lizards, and snakes (PETA, 2018). Lastly, to support and raise awareness of animal and environmental issues, many consumers have turned to vegan and imitation leather products as an alternative for genuine leather.

Issues with Plastic-based Synthetic Leather

Imitation leather is a common substitute choice for animal leather, it is also known as synthetic leather. Synthetic leather is more cost-effective in production and it is made with highly processed raw materials. The two most commonly researched types of synthetic leather are those produced with polyvinyl chloride (PVC) or polyurethane. Both are thermoplastic materials that can be shaped into a variety of forms (Medical Design Briefs, 2015). However, thermoplastics are not easily biodegradable, so they last for a long period of time.

Polyurethane is composed of isocyanates, polyester polyols, and additives. The two commonly used types of isocyanates are methylene diphenyl diisocyanate (MDI) and toluene diisocyanates (TDI) (Lee, 2002). These are usually mixed with additives such as carbon or silica to strengthen the matter. Thermal stabilizers, plasticizers, fire retardants, UV stabilizers, colorants, matting agents, pacifiers, or luster additives are also often added to the polymers and can raise the possibility of adverse health issues (Scalenghe, 2018). In addition, since polyurethanes have an unstable sensibility to sunlight, UV radiation could promote harmful reactions within the thermoplastic (Newman and Forciniti, 2001, Scalenghe, 2018).

Although synthetic fiber is free of animal cruelty, it can have a negative impact on humans and bring contaminations to the environment. Isocyanates are severe irritants to the mucous membranes of the eyes and gastrointestinal and respiratory tracts. Direct skin contact can also cause marked inflammation. Workers who are in contact with isocyanates make them subject to severe asthma attacks if they experience frequent exposure. There is evidence that both respiratory and dermal exposures can lead to

sensitization and death has been reported from severe asthma in some sensitized subjects (NIOSH, 2018). Some common symptoms are recurring eye irritation, nasal congestion, dry or sore throat, cold-like symptoms, cough, or shortness of breath. Workers who are constantly exposed to isocyanates should take safety precautions as the chemicals can trigger many health hazards

Rising Vegan Leather Industry

The vegan leather industry is on the rise because there is an increasing population of vegans. The number of global consumers who identify as vegan has grown 61% from 2014 to 2017 (Global Data, 2018). Veganism is referred as the movement or philosophy that advocates and entails abstinence from consuming any animal product - such as meat, dairy, eggs, leather, silk, etc. - and stands against widespread animal exploitation It is a growing movement that specifically aims at reinforcing the philosophy behind social justice through extending the morality towards animals, respecting their lives, and fostering compassion between animals (Ulusoy, 2015). Sustainability has become a key element across all industries because consumers, especially in the west, have become more aware about the practices surrounding goods they buy and have a keen interest of the origins of the goods (Choudhar, 2017). Therefore, the vegan leather industry is expected to rise with the evolution of veganism. According to Grand View Research, the vegan leather industry is expected to reach $85 billion by 2025. With the evolving textile technology, consumers are preferring vegan fashion, which refers to adopting non-leather products (Grand View Research, 2019). It is predicted that vegan leather will serve as the most suitable substitute for animal leather. Among materials that are being used to develop vegan leather are mangoes, apple, pineapple, and kombucha. This project aims to further increase this list by using banana peel reinforced with fibers from snake plant.

Vegan Leather in the Philippines

The country is not new with the idea of vegan leather production. In fact, Philippines is the source of pineapple leaves used by a London-based foreign company, Ananas Anam, in the production of Piñatex. Apart from that, no other vegan leather production is known in the country.

Mango Leather

The production of mango vegan leather is seen by the researcher as the inspiration for the development of banana (*Musa acuminata* × *balbisiana* var. Cardaba) peel vegan leather. With almost similar consistency when pulped, banana peel is at actual advantage with higher fiber content than of mango. In comparison using pectin content, a gelling and thickening agent necessary in production, banana peel has 13.0 – 21.7g/100g content (Wachirasiri, et al., 2009) whereas mango has 8.2g/100g content (Maldonado-Celis, et al., 2019).

Banana Peel Waste

Cardaba banana (*Musa acuminata* × *balbisiana* var. Cardaba) is a triploid hybrid (ABB) banana cultivar originating from the Philippines. This is the most popular variety for processing of banana chips in the domestic and export markets (Cruz et al., 2008) and is considered to be the most important variety of banana in the country (Tanaid and Lauzon, 2018). Banana fruit peels comprise a significant quantity of wastes produced from banana processing, which is equivalent to 40% of the total weight of fresh banana (Castillo-Israel et al., 2015). Many studies have been conducted in utilizing banana peels such as processing into bioethanol (Shaha et al., 2013) and citric acid (Kareem, S.O. & Rahman, R.A., 2013) Moreover, there are also a number of literatures that processes these peels into bioplastic (Galentsios et al., 2017, Sultan et al., 2017, & Azieyanti el al., 2020). The development of the peel into a vegan leather is yet to be explored.

Snake Plant Fiber

Snake plant (*Sansevieria trifasciata*) are species of plant in the family *Asparagaceae* that are highly abundant and are usually used are both indoor and outdoor ornament. Extracted fiber from this plant using water retting method showed good strength and fineness with low elongation (Kant & Alagh, 2015). In the study by Apalla (2021), the fabric made fiber from snake plant is comparable in quality to what is extracted from other sources or materials like piña fabric, corn husk fabric, water hyacinth fabric, coconut husk fabric and other native sources. Incorporating this fiber in the development of vegan leather from banana peels could provide reinforcement in terms of the material's mechanical properties.

(9) METHODOLOGY

Raw Material Sourcing and Study Sites

Discarded banana (*Musa acuminata* × *balbisiana* var. Cardaba) peels will be collected from an organized women's group from Burabod, Lagonoy, Camarines Sur that develops local banana food products. To validate that Cardaba variety is used for the vegan leather processing, samples will be sent to Jose Vera Santos Memorial Herbarium, University of the Philippines for identification and certification. Snake plants (*Sansevieria trifasciata*) will be collected from areas of San Jose, Camarines Sur.

Preparation of Raw Materials

Banana Peels

Collected banana peels will be transferred to the laboratory for quality assurance. After cutting off the stem end and the tip, banana peels are washed with water to remove soil and other adhering particles. Afterwards, it will be blended until homogenous and is set-up for vegan leather fabrication.

Snake Plant Fibers

Fibers from snake plant will be extracted using an accelerated retting method formulated by Abdullah et al. (2020) wherein fibers are obtained in 48-72 hours. In this process, hammered samples of snake plant were mixed with sodium hydroxide (1-2% of sample weight), molasses (1% of sample weight) and water (20 parts of sample) for 2-3 days with occasional stirring.

Fabrication of Vegan Leather

The fabrication of vegan leather using banana peels and snake plant fiber as reinforcement shall entail a novel method since no literature is available that explicitly provides the process of developing a similar product. Initial processing of the vegan leather shall be done by mixing the banana peel and snake plant fiber (50% w/w) with glycerin (30% w/w) as plasticizer (Arham et al., 2016), sodium alginate (10% w/w) as gelling agent and emulsifier (Rinaudo, 2014 & Sharma et al., 2018), calcium chloride (5% w/w) as quality enhancer for elongation (Zhang et al., 2020) and additives (5% w/w), and dehydrating it while being casted into a carrier cloth. In addition, banana peel has natural pectin content (Lee et al., 2010 & Kamble et al., 2017) that also contribute as gelling agent. Mechanical rolling and/or pressing will be done to ensure that the product is formed into sheets. Natural wax coating will also be incorporated. Coloring and other additives may be added to ensure the quality of the product.

Technical Evaluation of the Developed Vegan Leather

The technical performance of the fabricated vegan leather shall be evaluated using ISO standardized test used by Meyer et al. (2021) in the study that compares the technical performance of leather, artificial leather, and trendy alternatives. In addition, the characteristics of the developed vegan leather will be compared with locally available commercial vegan leather sourced from Bloomtex Fabric Warehouse (Polyurethane-based) and Jacinto and Lirio (Handmade) using same test procedure. Tests will be conducted trough third-party testing at Physical and Performance Testing Laboratory of Industrial Technology Development Institute and/or SGS Philippines Inc. - Multi-Laboratory (accredited by DTI-Philippine Accreditation Bureau for testing of leather and leather products).

Tensile Strength (ISO 3376)

Following the standardized process for determining of tensile strength of the leather (ISO 3376), the jaws of the tensile testing apparatus will be set (50 ± 1) mm apart if using the standard test piece or (100 ± 1) mm apart if using the large test piece. The test piece will be clamped in the jaws so that the edges of the jaws lie along the lines AB and CD. When the test piece is clamped, grain surface is ensured to lie in one plane. The length axis shall be parallel to the traction direction.

Afterwards, the machine will be run until the test piece breaks and record the highest force exerted as maximum force, F_{max}.

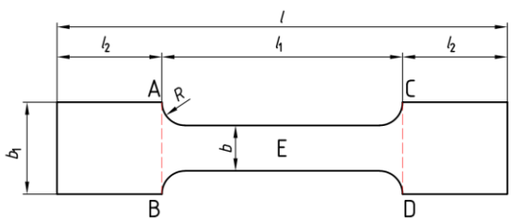


Fig. 2. Illustration of Test Piece for Tensile Strength

The tensile strength, T_n, in MPa (or newtons per square millimetre, if required) shall be calculated as:

$$T_n = \frac{F_{max}}{\bar{w} \cdot \bar{t}} \quad (1)$$

Where F_{max} is the highest force recorded (in newtons), $\bar{w}$ is the mean width of the test piece (in millimeters), and $\bar{t}$ the mean thickness of the test piece (in millimeters). Target value of tensile stength is >15N/mm^2 following chrome tanned upper leather for shoes (ISO 20942).

Tear Resistance (ISO 3377-1)

Following the standardized process for determining of tear resistance of the leather (ISO 3377-1), the jaws of the tensile testing machine will be set so that the clamps are at a suitable distance to

firmly hold the test piece. One leg of the test piece will be clamped in one jaw of the tensile test machine. The other leg will be folded through 180° and will be clamped in the other jaw. The long edges of the test piece will be ensured to be parallel to the testing direction of the machine.

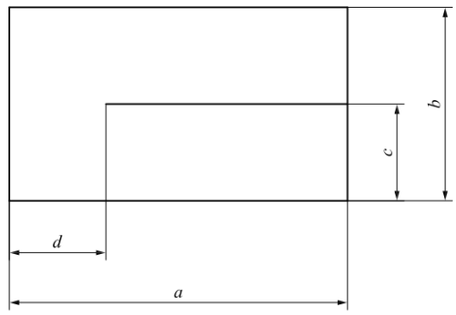


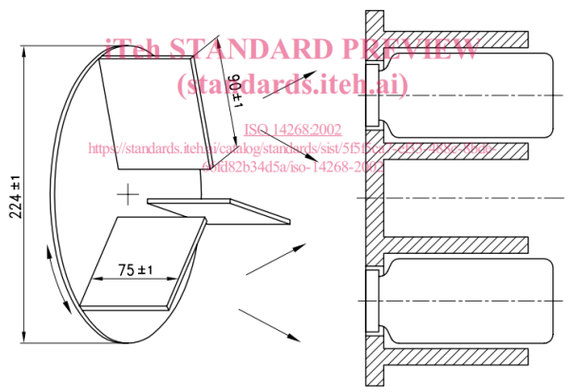
Fig. 3. Illustration of Test Piece for Tear Resistance for Single Edge Tear

The tensile test machine will be run until the test piece is torn apart and record the force-extension plot. The tear load of the test piece is determined as the arithmetic mean of the forces on the peak trace, proceeding in the following way.

Water Vapor Permeability (ISO 14268)
Following the standardized process for determining of water vapor permeability of the leather (ISO 14268), a container first be half-filled with freshly regenerated silica gel. A test piece will then be placed centrally over the open container so that the surface which is exposed to the higher humidity when the final product is in use is uppermost. Screw top will be fitted to the container and tighten so that the test piece is securely held around the edge and the container is sealed. The container is placed on a turntable and is started.

Using vernier callipers the internal diameter of the neck of a second container will be measured (to the nearest 0,1 mm) in two mutually perpendicular directions and calculate the mean diameter. After 20 h ± 4 h, the first container is removed from the test machine then, as rapidly as possible, the second container will be half-filled with freshly regenerated silica gel, the test piece will be removed and will be screwed top from the first container, place on the second container. The mass will then be recorded (M_0).

Afterwards, the container on the vertical turntable will replaced and the test machine will be started. After 11.5 h ± 4.5 h the test machine will be stopped. and note the time. The container will be removed and be reweighed. Mass will be recorded (M_1).



iTeh STANDARD PREVIEW
(standards.iteh.ai)
ISO 14268:2002
https://standards.iteh.ai/catalog/standards/sist/5f5b...
...d82b34d5a/iso-14268-2002

Fig. 4. General arrangement of test machine (all dimensions in millimetres)

The water vapour permeability, P_{wv}, in milligrams per square centimetre hour, will be calculated using the following equation:

$$P_{wv} = \frac{7639M}{d^2t} \quad (2)$$

Where M is the increase in mass of the container (M_1 - M_0, in milligrams), d is the mean diameter of the neck of the container (in millimeters), and t is the time between the first and second weighing in (minutes). >>0.8 mg/(cm²•h) is targeted following chrome tanned upper leather for shoes (ISO 20942).

Water Vapor Absorption (ISO 17229)

Following the standardized process for determining of water vapor absorption of the leather (ISO 17229), the internal diameter of a cylindrical container will be measured using vernier calipers (to the nearest0,1 mm) in two mutually perpendicular directions and the mean diameter will be calculated. 50 ml ± 5 ml of distilled or deionized water will be poured at 20 °C ± 2 °C or 23 °C ± 2 °C into the cylindrical metal container. The test piece will be placed centrally over the container with the side which would be exposed to the higher humidity in use facing downwards. A disc of impermeable material is placed over the test piece and the upper ring will be clamped. The container will be kept at a temperature of 20 °C ± 2 °C or 23 °C ± 2 °C for 8 h ± 0.1 h. Test piece will be weighed immediately to the nearest 0.001 g and the mass will be recorded.

The water vapour absorption, A_{wv}, in milligrams per square centimetre, will be calculated using the following equation:

$$A_{wv} = \frac{4(m_2 - m_1) \times 10^5}{\pi d^2} \quad (3)$$

Where m_1 is the initial mass of the test piece (in grams), m_2 is the final mass of the test piece, in grams, and d is the internal diameter of the cylindrical container (in millimeters).

Flex Resistance (ISO 32100)
Following the standardized process for determining of flex resistance of the leather (ISO 3376), flexometer method will be employed. In this direct and machine-centered method, test piece will be folded with the surface to be tested inwards and will be clamped in an upper movable clamp and with the surface to be tested outwards in a lower fixed clamp. Movement of the upper clam caused a fold in the test piece to run along with it. The test piece will be examined periodically for damage. >80.000 flex cycles is targeted following chrome tanned upper leather for shoes (ISO 20942).

Morphological Analysis
Uniform dispersion of all components in a given matrix is highly necessary to achieve stable adhesion and good mechanical properties of a leather (Kong, 2016). The morphologies of the polymeric matrix of the developed vegan leather will be evaluated using field-emission scanning electron microscopy (FESEM) at Advanced Device and Materials Testing Laboratory (ADMATEL) of DOST-Industrial Technology Development Institute. Moreover, porosity will be evaluated at Physical and Performance Testing Laboratory of the same institute.

Production of Representative Test Products from the Developed Vegan Leather
Test products which may include shoes, tote bag, purse, and/or wallet shall be developed once the optimum process for the fabrication of the vegan leather has been achieved. Due to the nature of the project, the products are only representative samples that could support further use of the vegan leather in the industry.

(10) TECHNOLOGY ROADMAP (if applicable) (use the attached sheet)

Not applicable to the study.

(11) EXPECTED OUTPUTS (6Ps)

Publication	One (1) research in related international-refereed/ISI-refereed Journal.
Patent/Intellectual Property	One (1) copyright certificate issued by the National Library of the Philippines or the Intellectual Property Office of the Philippines (IPOPhil) for the manuscript produced. One (1) patent/utility model application to the Intellectual Property Office of the Philippines (IPOPhil) for the technology developed.
Products	At least two (2) representative test products made out of vegan leather from banana peels reinforced with snake plant fiber.
People Service	Training of one (1) project staff, hiring of one (1) science aide and training of members of one (1) cooperative adopter.
Place and Partnership	Lagonoy Women's Cooperative, Burabod, Lagonoy, Camarines Sur/Local Government Unit of Lagonoy (Target cooperative adopter of technology). Technology will be transferred to said cooperative and members will be trained to fabricate the vegan leather, as well as, develop products from it which will then be sold as source of livelihood. Jacinto and Lira (Target private company adopter)

Policy	Policy proposal for standards and guidelines for the development of vegan leather from banana peels reinforced with snake plant fiber, and be used as basis into developing similar products.

(12) POTENTIAL OUTCOMES

The produced vegan leather from banana (*Musa acuminata* × *balbisiana* var. Cardaba) peels reinforced with snake plant (*Sansevieria trifasciata*) fiber could be used in the development of multiple products that usually uses leather and plastic-based faux leather. Result of this proposed project, both the process and the product, could directly be applied in the expanding textile industry, particularly in the vegan leather line where no product of the same kind is produced. With this, it could open a new industry of banana peel vegan leather that could be pitted against kombucha leather, mango leather, and pinatex – a vegan leather from pineapple textile. Not only it creates another sustainable and environmentally-sound product but opens potential livelihood that could help alleviate poverty in the community.

(13) POTENTIAL IMPACTS (2Is)

Social Impact
With an increase in ethical awareness, people have begun to criticize the unethical issues associated with the use of animal materials. In a way, the project could provide awareness toward vegan materials and the relationship between the interest in ethical subjects such as animals, the environment, and vegan materials.

Economic Impact
The proposed vegan leather utilizes raw materials that are highly accessible. Banana (*Musa acuminata* × *balbisiana* var. Cardaba) peel is considered as waste of banana processing, whereas snake plant (*Sansevieria trifasciata*), where the fiber to be used as reinforcement is derived, is naturally abundant and is easy to propagate. The project could open another competitor in the vegan leather industry and further provide additional livelihood opportunity.

(14) TARGET BENEFICIARIES

Community. The technology developed, when transferred to the community, could provide livelihood from the creation of vegan leather products such as wallet, bags, and the likes.

Banana Industry. The project valorizes banana (*Musa acuminata* × *balbisiana* var. Cardaba) peel, a known agricultural waste from banana production.

Researchers. Results obtained from the project could be used in further improving the product. Moreover, with the establishment of a process of developing vegan leather from banana peel reinforced with fiber from snake plant, the project could also pave way in the development of other waste-sourced vegan leather.

(15) SUSTAINABILITY PLAN (if applicable)

Not applicable to the study.

(16) GENDER AND DEVELOPMENT (GAD) SCORE (refer to the attached GAD checklist)

10.0

(17) LIMITATIONS OF THE PROJECT

The proposed project only limits itself to the production of vegan leather using banana (*Musa acuminata* × *balbisiana* var. Cardaba) peel reinforced with snake plant fiber (*Sansevieria trifasciata*), development of the fabrication process, evaluation of its quality, and construction of test vegan leather products. Production at an economic standpoint and marketing-related study such as product feasibility is not included in the project

(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN (List possible risks and assumptions in attaining target outputs or objectives.)

Risks & Assumptions	*Action Plan*
Late receipt of requested equipment and reagents.	Early sending of request for price quotation to prospect suppliers and prompt response to queries concerning procurement. Moreover, closely working with the procurement office to expedite the process.

Inaccessible raw materials (Banana peel and snake plant)	Banana are available year-round and peels could be requested from banana vendors. Snake plant, on the other hand, is highly abundant in the area where the project will be conducted. In addition, snake plant is easy to propagate.
Unavailable testing materials for quality evaluation.	Contacting other laboratories and laboratory service providers to conduct quality evaluation of the product.

(19) LITERATURE CITED

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(20) PERSONNEL REQUIREMENT

Position	Percent Time Devoted to the Project	Responsibilities
Project Leader	100%	Implement the project. Lead in all the fabrication and quality evaluation. Collate all data and prepare the terminal report.
Project Staff	50%	In-charge in the development of product using fabricated vegan leather. Assist in the sourcing of materials for test product development.
Research Assistant (Science Aide)	100%	Assist in all activities of the project. Prepare documents for reports. Communicate with administrative matters.

(21) BUDGET BY IMPLEMENTING AGENCY

IMPLEMENTING AGENCY	PS	MOOE	EO	Total
Year 1	-	80,000.00	70,000.00	150,000.00
TOTAL	**-**	**80,000.00**	**70,000.00**	**150,000.00**

(22) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: 1 (number)

Title of the Project	Funding Agency	Involvement in the Project
Phytochemical Analysis of Select Plants used For Traditional Herbal Medicine in Partido District, Camarines Sur, Philippines	Partido State University	Project Leader

(23) OTHER SUPPORTING DOCUMENTS (Please refer to page 2 for the additional necessary documents.)

I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled is within the prescribed number as stipulated in the DOST-GIA Guidelines. Any

willful omission/false statement shall be a basis of disapproval and cancellation of the project.

	SUBMITTED BY (Project Leader)	ENDORSED BY (Head of the Agency)
Signature		
Printed Name	**Kier P. Gasga, RChE, RChT**	**Raul G. Bradecina, PhD**
Designation/Title	Project Leader	SUC President III
Date	27 January 2022	27 January 2022

Note: See guidelines/definitions at the back.

DOST Form 2 (for Basic/Applied Research)
DETAILED R & D PROJECT PROPOSAL

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the detailed R&D proposal for the component project together with the detailed proposal of the whole Program, project workplan, line-item budget (LIB), 1-page curriculum vitae of the Project Leader, and Certificate of Incorporation or DTI Registration (if applicable) and other applicable supporting documents required under item II.23 below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

II. Operational Definition of Terms:

1. Title- the identification of the Program and the component projects.

Project- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

Project Leader- refers to a project's principal researcher/implementer.

Project Duration- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

Implementing Agency- the primary organization involved in the execution of a program/project which can be a public or private entity

2. Cooperating Agency/ies- refers to the agency/ies that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

3. Site/s of Implementation- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

4. Type of Research- indicates whether the project is basic or applied.
Basic research- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.
Applied research- is an investigation undertaken in order to utilize data/information gathered from fundamental/basic researches or to acquire new knowledge directed primarily towards a specific practical aim or objective with direct benefit to society.

5. R&D Priority Area and Program- based on the Harmonized National R&D Agenda 2017-2022, indicates which R&D agenda the project can be categorized in: Agriculture, Aquaculture and Natural Resources; Health; Industry, Energy, and Emerging Technology; Disaster Risk Reduction and Climate Change Adaptation; and Basic Research. Indicate also the specific Commodity/Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure.

Sustainable Development Goal (SDG) Addressed- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

6. Executive Summary- briefly discusses what the whole proposal is about

7. Introduction- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

7.1. Rationale- brief analysis of the problems identified related to the project
Significance- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

7.2. Scientific Basis- other scientific findings, conclusions or assumptions used as justification for the research
Theoretical Framework- the structure that summarizes concepts and theories that serve as basis for the data analysis and interpretation of the research data.

7.3. Objectives- statements of the general and specific purposes to address the problem areas of the project.

8. Review of Literature- refers to the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific/technical merit; (c) results of related research conducted by the same Project Leader, if any; (d) Prior Art Search, and; (e) other relevant materials.

9. Methodology- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

10. Technology Roadmap (if applicable)**-** a visual document that communicates the plan for technology. It is a flexible planning technique to support strategic and long-range planning by matching short- and long-term goals to specific technology solutions.

11. Expected Outputs (6Ps)- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

Publication- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.
Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.
Product- invention with a potential for commercialization.
People Service- people or groups of people, who receive technical knowledge and training.
Place and Partnership- linkage forged because of the study.
Policy- science-based policy crafted and adopted by the government or academe as a result of the study.

12. Potential Outcomes- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

13. Potential Impacts

Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.
Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

14. Target Beneficiaries- refers to groups/persons who will be positively affected by the conduct of the project.

15. Sustainability plan- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

16. Gender and Development (GAD) Score- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, “Women in Development and Nation Building Act,” interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

17. Limitations of the Project- refer to restrictions or constraints in the conduct of the project.

18. Risk- refers to an uncertain event or condition that its occurrence has a negative effect on the project.
Assumption- refers to an event or circumstance that its occurrence will lead to the success of the project.

19. Literature Cited- an alphabetical list of reference materials (books, journals and others) reviewed. Use standard system for citation.

20. Personnel Requirement- details on the position of personnel to be involved in the project, percent time devoted to the project, and responsibilities.

21. Budget By Implementing Agency- personnel services (PS), maintenance and other operating expenses (MOOE), and equipment outlay (EO) requirement of the project by implementing agency for

Year 1 and for the whole duration of the project. Please refer to the DOST-GIA Guidelines for the details (Section IX.B of DOST Administrative Order (A.O.) 011, s. 2020).

a. **PS**- total requirement for wages, salaries, honoraria, additional hire and other personnel benefits.

b. **MOOE**- total requirement for supplies and materials, travel expenses, communication, and other services.

c. **EO**- total requirement for facilities and equipment needed by the Program.

22. Other Ongoing Projects Being Handled By the Project Leader- list of ongoing projects being handled by the Project Leader funded by the DOST-GIA Program and other sources, and the accompanying responsibilities relevant to the project.

23. Other supporting documents required- as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program:

a. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]
b. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]
c. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]
d. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]
e. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);
f. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and
g. For the private non-profit/non-government/people's organizations and startups:
    i. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority (CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;
    ii. Co-signers Statement (if applicable);
    iii. Copy of latest Income Tax Return;
    iv. Mayor's permit where the business is located;
    v. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);
    vi. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;
    vii. Disclosure of other related business, if any;
    viii. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;
    ix. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;
h. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and
i. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

III. Criteria for Evaluation:

A. Criteria for Evaluating Proposals

Criterion	Definition
Relevance or Significance	Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development
Technical / Scientific Merit	Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner
Budget Appropriateness	The proposed budget is commensurate to the proposed work plan and deliverables.
Competence of Proponent	Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget.

B. Governing Council / Board and EXECOM's Evaluation Criteria

Criteria	Indicators	Raw Score
A. Soundness of Proposal (20%)	R&D addresses relevant sectoral need (applicable to pressing concern)	5
	Solution provided is most effective (compared to other proposed solutions)	5
	Proposed budget is reasonable (project is not expensive vis-a-vis output)	5
	Work plan is doable in a given timeframe	5
B. Suitability of Output (30%)	R&D output is cost-effective (cost is competitive in relation to new or existing products or process)	5
	Has identified partners to adopt the technology (with letter of support from the head of the company)	5
	Output can be commercialized (through an existing manufacturer, spin-off or start-up company)	5
	R&D utilization is timely (output should not be overtaken by other solutions)	5
C. Significance of Outcome (30%)	Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI)	5
	Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity	5
	Environment: enhanced environmental health standards, no adverse effect to the environment	5
	Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project)	5
D. Competence of Proponent (20%)	Proponent's expertise aligned with the proposal	5
	Collaboration with relevant agencies and/or industry partners	5
	Thorough understanding of the proposal's deliverables	5
	DOST has good experience with the proponent	5

C. Additional Criteria on Gender and Development (GAD)